Vakint reduction benchmark

What vakint computes exactly or numerically, and to what precision · RustRed versus alphaLoop, MATAD and FMFT · reduction graphs from one to five loops

2026-11-07 · AMD EPYC 9754 128-Core Processor · one core per timed process except 10 marked cells · vakint 0.2.0

What vakint computes today

RELEASED data tree vakint-external-data v0.2.2 (0faa1c795664) · 5L basis BASIS-v2 · tables 7c9148d895cb5b0e (final) · generated 2026-11-11T03:37:30Z
Only what this data tree ships. Results derived since the release (R operation, new basis masters, adopted PSLQ forms) are in the PREVIEW edition of this document.

For every single-scale vacuum integral from one to five loops: which Laurent coefficients vakint returns **exactly** (S: in constants it evaluates at any precision; S*(X): in constants known to X digits), which only **numerically** (N(X): X validated digits), and which are **structural zeros**. (a) counts the master coefficients, (c) states what a user gets for an arbitrary L-loop input, (d) lists the constants, (e) says how the forms were obtained and checked. The per-master table (b) is precision table 1 (section "Precision of the masters and of the results").

S exact, periods at any precision **0** structural zero **S*(X)** exact, periods to X digits
N(X) numerical only, X digits[1] — not available

[1] Every published digit figure of a numerical value (N(X), numerical payloads, finite literals) is its validated claim minus one digit (user ruling 2026-11-11), which absorbs discrepancies of 0.3–1 digit between a symbolic form and the stored claim. The lifted constants behind S*(X) keep their own digits.

(a) Master coefficients per loop and Laurent order — number of master coefficients in each class (held: masters holding the order); S*: range of the digits of the limiting constants; N: fewest and median validated digits (rounded down).

loops	class	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}	total
1L 1 masters	held	—	—	—	—	1	1	1	1	1	1	1	
	S					**1**	**1**	**1**	**1**	**1**	**1**	**1**	**7**
2L 2 masters	held	—	—	—	2	2	2	2	2	2	1	—	
	S				**2**	**2**	**2**	**2**	**2**	**2**	**1**		**13**
3L 38 masters	held	—	—	38	38	38	38	38	38	21	3	—	
	S			**37**	**37**	**38**	**38**	**38**	**38**	**21**	**3**		**250**
	0			**1**	**1**	·	·	·	·	·	·		**2**
4L 74 masters	held	—	74	74	74	74	74	74	20	4	—	—	
	S		**53**	**57**	**64**	**67**	**28**	**28**	**20**	**4**			**321**
	0		**21**	**17**	**10**	**7**	·	·	·	·			**55**
	S*(X)		·	·	·	·	**46** 2395–20000 d	**46** 2348–20000 d	·	·			**92**
5L 131 masters basis BASIS-v2	held	131	131	131	131	131	131	—	—	—	—	—	
	S	**53**	**59**	**69**	**57**	**22**	**11**						**271**
	0	**78**	**72**	**60**	**51**	**30**	·						**291**
	S*(X)	·	·	·	·	**19** 2395–20000 d	**10** 2348–20000 d						**29**
	N(X)	·	·	**2** ≥47, med 55.5	**23** ≥46, med 64	**60** ≥34, med 56	**110** ≥2, med 56						**195**

(c) Result of any single-scale vacuum input with L loops — worst case over the admitted inputs (RustRed backend), spurious poles of the reduction and lower-loop factors of 5L products included (letters: the notes of precision table 2). "Not every input": some inputs need master orders the data do not hold and get MissingMasterOrders (no value, never a wrong one).

input	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
1L					**exact**	**exact**	**exact**	**exact**	**exact**	**exact**	**exact**
2L				**exact**	**exact**	**exact**	**exact**	**exact**	**exact**	not every input[b] others S	—
3L			**exact**	**exact**	**exact**	**exact**	**exact**	not every input[c] others S	not every input[d] others S	not every input[e] others S	—
4L		**exact**	**exact**	**exact**	**exact** guaranteed[2f]	**exact S*** 2395 d[g]	not every input[h] others S*(2348)	not every input[i] others S	not every input[j] others S	—	—
5L	**exact**	**N** ≥ 64 d others S[m]	**N** ≥ 46 d others N(47)[n]	not every input[o] others N(46)	not every input[p] others N(34)	not every input[q] others N(2)	—	—	—	—	—

[2] **4L ε⁻¹ guaranteed exact** (R-operation): the pole coefficients of every single-scale vacuum integral are free of the 4L-only constants; verified exactly on vakint's own reduction data: Verified exactly (coefficient of every 4L/constants.json constant at ε⁻⁴..ε⁻¹ = 0): all 286527 records of the shipped 4L caches (77724 with spurious poles, 1375 with 1/ε²), 2230 sampled keys of track LL and 514 fresh keys outside both (all four families, up to 6 dots and 4 numerator powers); 338832 pole x constant

products cancel. The spurious-pole residue vectors span dimension x 6, h 5, bmw 5, fg 3 and every basis vector has zero 4L-constant content. vakint itself returns these orders beyond 20000 digits (4L/P15 + dot: ε^{-1} to 20524 d in Symbolic, 20499 d of 20500 requested in Numerical mode; finite inputs P15, P17, P18 + dot: exact 0). So the K5Moment(1), K5Moment(3) terms that spurious poles bring in through 4L/P6 ε^0, 4L/P7 ε^0, 4L/P11 ε^0 cancel identically, and vakint already returns this order free of them. Proof: Proof: by the R-operation, K G = -Delta(G) - K sum_S Delta(S) G/S for any 4-loop single-scale (all lines of mass m) vacuum integral G with polynomial numerators; the counterterms Delta are polynomials in m^2 with pole coefficients in multiple zeta values (infrared rearrangement onto massless propagators of <= 3 loops), and the quotients G/S (<= 3 loops) enter only through ε^(3-L_q) (3L ε^0, 2L ε^1, 1L ε^2), all S (polylogarithms at sixth roots of unity, no 4L constant). So every pole coefficient of G is free of the 4L constants; vakint's exact assembly shows the coefficient of each one to be identically zero (exact rational arithmetic) on every record checked.

(e) How the symbolic forms were obtained and checked

Closed forms. 1L-3L and the poles of 4L: closed forms from the literature and the MATAD/FMFT expansions (harmonic polylogarithms at the sixth root of unity, the MATAD constants), carried exactly by vakint. 4L finite parts: exact in 50 lifted four-loop constants (Bessel moments, two-dimensional ε-jets) computed to 2348-20000 digits (S*). Some S rows rest on numerical identities of the constant basis checked to thousands of digits (8 identities; PRECISION-TABLES.md).

Five loops. Structural zeros from the UV forest bound and exact rows from pole cancellation; 196 of the 591 exact 5L orders are PSLQ identifications (9 S*) and 4 measured zeros, each passing the 40-excess-digit bar (numerical identifications, not proofs). The other 5L orders are numerical (high-precision grids and dispersive gluing; digits = validated significant digits).

R operation (preview). Pole coefficient K G = (A) − Δ: part (A) = −K Σ_S Δ(S) G/S from the subdivergences (ordinary massive R operation, exact, lower-loop coefficients of vakint); Δ = overall counterterm, a pure number. Evidence classes: exact derivation (ω < 0: no overall counterterm, the result is complete); confirmed by R* (Δ derived by R*); fitted counterterm (Δ found as an integer relation of [numeric − (A)] in 1, ζ_3, π^4, ζ_5, π^6, ζ_3^2, ζ_7, $\zeta_3\pi^4$, with its margin in digits). Fits below the acceptance margin are listed as marginal and not applied; poles whose Δ the fit did not identify, or without a part (A) yet, stay numerical and are listed.

4L poles. From ε^-1 down the 4L result is exact and guaranteed: by the R-operation its pole coefficients are free of the 4L-only constants, verified exactly on vakint's reduction data (footnote 2).

Numerical check. Every symbolic form is compared with the numerical value it replaces, to that value's claimed digits (user ruling 2026-11-10), at adoption and again when the release tooling integrates it; a mismatch blocks the substitution. Published digits of numerical values are the claim minus one digit (user ruling 2026-11-11), so discrepancies of up to one digit are absorbed; larger ones are flagged and listed.

(d) Named constants — every constant in the exact coefficients of the data tree. Built-ins are evaluated by vakint at any requested precision; finite records (4L/constants.json) are known to the digits shown, which bound every S^* coefficient that contains them. Traditional forms (periods/l6 dictionary): verified to the full stored digits; not yet adopted in the data full formulas and the explicit integral definitions in the appendix below and in `docs/benchmark/data/CONSTANTS-TRADITIONAL.md`. D5bar3 is undecided.

constant	definition	digits	any precision	coefficients (released)	traditional form (verified, not yet adopted)
`cl2`	$Cl_2(\pi/3) = \mathrm{Im}\,Li_2(e^{i\pi/3})$	any	yes	391 (2L 5, 3L 93, 4L 179, 5L 114)	Cl2(π/3) (agrees to 20004.3 d with the stored 20000 d)
`cl4`	$Cl_4(\pi/3) = \mathrm{Im}\,Li_4(e^{i\pi/3})$	any	yes	202 (2L 3, 3L 54, 4L 108, 5L 37)	Cl4(π/3) (agrees to 20005.8 d with the stored 20000 d)
`D5bar1`	MATAD D5bar(1): ε^1 coefficient of the three-loop tetrahedron with five massive lines	any	yes	83 (3L 12, 4L 53, 5L 18)	17 terms in ζ, ln, π, classical Li_n, Clausen, Im/Re Li_n(i/√3), multiple polylogs (agrees to 20008.7 d with the stored 19998 d)
`D5bar2`	MATAD D5bar(2): ε^2 coefficient of the three-loop tetrahedron with five massive lines	any	yes	39 (3L 6, 4L 24, 5L 9)	111 terms in ζ, ln, π, classical Li_n, Clausen, Im/Re Li_n(i/√3), log-sine, multiple polylogs (agrees to 20008.0 d with the stored 19998 d)
`D6bar1`	MATAD D6bar(1): ε^1 coefficient of the three-loop tetrahedron with six massive lines	any	yes	16 (3L 2, 4L 8, 5L 6)	34 terms in ζ, ln, π, classical Li_n, Clausen, Im/Re Li_n(i/√3), multiple polylogs (agrees to 20008.4 d with the stored 19998 d)
`D6bar2`	MATAD D6bar(2): ε^2 coefficient of the three-loop tetrahedron with six massive lines	any	yes	8 (3L 1, 4L 4, 5L 3)	118 terms in ζ, ln, π, classical Li_n, Clausen, Im/Re Li_n(i/√3), log-sine, multiple polylogs (agrees to 20008.0 d with the stored 19998 d)
`GPLImag`	Im G($a_1,\ldots,a_n$; 1): multiple polylogarithms, letters 0, ±1, sixth roots of unity, weight ≤ 6	any	yes	118 (2L 2, 3L 36, 4L 63, 5L 17)	—
`GPLReal`	Re G($a_1,\ldots,a_n$; 1): multiple polylogarithms, letters 0, ±1, sixth roots of unity, weight ≤ 6	any	yes	174 (2L 2, 3L 42, 4L 96, 5L 34)	—
`log(2)`	ln 2	any	yes	222 (2L 2, 3L 54, 4L 120, 5L 46)	—
`log(3)`	ln 3	any	yes	282 (2L 4, 3L 72, 4L 141, 5L 65)	—
`pi`	π	any	yes	633 (1L 5, 2L 9, 3L 175, 4L 289, 5L 155)	—
`polylog(4,1/2)`	Li_4(1/2)	any	yes	222 (2L 2, 3L 54, 4L 120, 5L 46)	—
`polylog(5,1/2)`	Li_5(1/2)	any	yes	79 (3L 12, 4L 51, 5L 16)	—
`polylog(6,1/2)`	Li_6(1/2)	any	yes	32 (3L 3, 4L 23, 5L 6)	—
`PolylogImag`	Im Li_n(z), $z = e^{i\pi/3}$ (n ≤ 6) or i/√3	any	yes	282 (2L 4, 3L 72, 4L 141, 5L 65)	—
`s6`	$s_6 = \Sigma_{m\geq k\geq 1} (-1)^{m+k}/(m^5\,k)$ (alternating Euler sum, weight 6)	any	yes	32 (3L 3, 4L 23, 5L 6)	8 terms in ζ, ln, π, classical Li_n (agrees to 20006.5 d with the stored 20000 d)
`T111bar4`	MATAD T111bar(4): ε^4 coefficient of the two-loop sunset with three massive lines	any	yes	44 (2L 1, 3L 18, 4L 19, 5L 6)	11 terms in ζ, ln, π, Clausen, Im/Re Li_n(i/√3), log-sine (agrees to 20003.9 d with the stored 19998 d)
`zeta(3)`	ζ(3)	any	yes	528 (1L 4, 2L 7, 3L 139, 4L 243, 5L 135)	—
`zeta(5)`	ζ(5)	any	yes	221 (1L 2, 2L 3, 3L 61, 4L 111, 5L 44)	—
`zeta(7)`	ζ(7)	any	yes	12 (3L 3, 4L 5, 5L 4)	—
`K5LogMoment(1)`	$3\int_0^\infty t^1\,K_0(t)^5\,(\ln t + \gamma_E - \ln 2)\,dt$ (log-weighted Bessel moment)	**20000**	no	91 (4L 65, 5L 26)	$3\int_0^\infty t^1\,K_0(t)^5\,(\ln t + \gamma_E - \ln 2)\,dt$
`K5LogMoment(3)`	$3\int_0^\infty t^3\,K_0(t)^5\,(\ln t + \gamma_E - \ln 2)\,dt$ (log-weighted Bessel moment)	**20000**	no	91 (4L 65, 5L 26)	$3\int_0^\infty t^3\,K_0(t)^5\,(\ln t + \gamma_E - \ln 2)\,dt$
`K5Moment(1)`	$c_{5,1} = \int_0^\infty t^1\,K_0(t)^5\,dt$ (Bessel moment, Broadhurst notation)	**20000**	no	121 (4L 92, 5L 29)	$c_{5,1} = \int_0^\infty t^1\,K_0(t)^5\,dt$ (Broadhurst)
`K5Moment(3)`	$c_{5,3} = \int_0^\infty t^3\,K_0(t)^5\,dt$ (Bessel moment, Broadhurst notation)	**20000**	no	117 (4L 88, 5L 29)	$c_{5,3} = \int_0^\infty t^3\,K_0(t)^5\,dt$ (Broadhurst)
`T111bar5`	MATAD T111bar(5): ε^5 coefficient of the two-loop sunset with three massive lines	**19997**	no	2 (4L 1, 5L 1)	14 terms in ζ, ln, π, Clausen, Im/Re Li_n(i/√3), log-sine (agrees to 20003.4 d with the stored 19998 d)
`BNbar5`	MATAD BNbar(5): ε^5 coefficient of the three-loop equal-mass banana	**19705**	no	6 (4L 4, 5L 2)	13 terms in ζ, ln, π, classical Li_n, multiple polylogs (agrees to 19712.3 d with the stored 19706 d)
`K5MomentTaylor(1,2)`	$[\varepsilon^2]\,e^{3(\gamma_E-\ln 2)\varepsilon}\int_0^\infty t^{1+3\varepsilon}\,K_\varepsilon(t)^5\,dt$ (ε-jet of the D = 2−2ε four-loop banana)	**19701**	no	53 (4L 39, 5L 14)	$[\varepsilon^2]$ of the D = 2−2ε four-loop banana in Bessel form ($c_{5,1}$ jet)
`K5MomentTaylor(1,3)`	$[\varepsilon^3]\,e^{3(\gamma_E-\ln 2)\varepsilon}\int_0^\infty t^{1+3\varepsilon}\,K_\varepsilon(t)^5\,dt$ (ε-jet of the D = 2−2ε four-loop banana)	**19701**	no	25 (4L 19, 5L 6)	$[\varepsilon^3]$ of the D = 2−2ε four-loop banana in Bessel form ($c_{5,1}$ jet)
`K5MomentTaylor(1,4)`	$[\varepsilon^4]\,e^{3(\gamma_E-\ln 2)\varepsilon}\int_0^\infty t^{1+3\varepsilon}\,K_\varepsilon(t)^5\,dt$ (ε-jet of the D = 2−2ε four-loop banana)	**19701**	no	4 (4L 3, 5L 1)	$[\varepsilon^4]$ of the D = 2−2ε four-loop banana in Bessel form ($c_{5,1}$ jet)
`K5MomentTaylor(3,2)`	$[\varepsilon^2]\,e^{3(\gamma_E-\ln 2)\varepsilon}\int_0^\infty t^{3+3\varepsilon}\,K_\varepsilon(t)^5\,dt$ (ε-jet of the D = 2−2ε four-loop banana)	**19690**	no	53 (4L 39, 5L 14)	$[\varepsilon^2]$ of the D = 2−2ε four-loop banana in Bessel form ($c_{5,3}$ jet)
`K5MomentTaylor(3,3)`	$[\varepsilon^3]\,e^{3(\gamma_E-\ln 2)\varepsilon}\int_0^\infty t^{3+3\varepsilon}\,K_\varepsilon(t)^5\,dt$ (ε-jet of the D = 2−2ε four-loop banana)	**19690**	no	25 (4L 19, 5L 6)	$[\varepsilon^3]$ of the D = 2−2ε four-loop banana in Bessel form ($c_{5,3}$ jet)
`K5MomentTaylor(3,4)`	$[\varepsilon^4]\,e^{3(\gamma_E-\ln 2)\varepsilon}\int_0^\infty t^{3+3\varepsilon}\,K_\varepsilon(t)^5\,dt$ (ε-jet of the D = 2−2ε four-loop banana)	**19690**	no	4 (4L 3, 5L 1)	$[\varepsilon^4]$ of the D = 2−2ε four-loop banana in Bessel form ($c_{5,3}$ jet)
`PR14ep2`	$[\varepsilon^2]$ of the FMFT four-loop master PR14 in D = 4−2ε (unit masses)	**9925**	no	9 (4L 4, 5L 5)	$[\varepsilon^2]$ of a 7-line four-loop vacuum integral in D = 4-2ε
`D5bar3`	MATAD D5bar(3): ε^3 coefficient of the three-loop tetrahedron with five massive lines	**9924**	no	6 (4L 4, 5L 2)	MATAD D5bar(3), weight 7: UNDECIDED (no relation found; a decisive test needs ~12000 strict digits)
`PR14ep3`	$[\varepsilon^3]$ of the FMFT four-loop master PR14 in D = 4−2ε (unit masses)	**9924**	no	2 (4L 1, 5L 1)	$[\varepsilon^3]$ of a 7-line four-loop vacuum integral in D = 4-2ε
`PR5D2Eps4`	$[\delta^4]\,e^{4\gamma_E\delta}$ × FMFT four-loop topology PR5 in D = 2−2δ, unit masses, Euclidean	**9923**	no	6 (4L 4, 5L 2)	$[\delta^4]$ of a 6-line four-loop vacuum integral in D = 2-2δ
`PR7D2`	FMFT four-loop topology PR7 in D = 2, unit masses, Euclidean	**5542**	no	86 (4L 61, 5L 25)	a 6-line four-loop vacuum integral in D = 2
`PR7D2Eps1`	$[\delta^1]\,e^{4\gamma_E\delta}$ × FMFT four-loop topology PR7 in D = 2−2δ, unit masses, Euclidean	**5540**	no	71 (4L 47, 5L 24)	$[\delta^1]$ of a 6-line four-loop vacuum integral in D = 2-2δ
`PR7D2Eps2`	$[\delta^2]\,e^{4\gamma_E\delta}$ × FMFT four-loop topology PR7 in D = 2−2δ, unit masses, Euclidean	**5538**	no	50 (4L 37, 5L 13)	$[\delta^2]$ of a 6-line four-loop vacuum integral in D = 2-2δ

constant	definition	digits	any precision	coefficients (released)	traditional form (verified, not yet adopted)
PR7D2Eps3	$[\delta^3]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR7 in D = 2−2δ, unit masses, Euclidean	**5538**	no	25 (4L 19, 5L 6)	$[\delta^3]$ of a 6-line four-loop vacuum integral in D = 2-2δ
PR7D2Eps4	$[\delta^4]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR7 in D = 2−2δ, unit masses, Euclidean	**5535**	no	4 (4L 3, 5L 1)	$[\delta^4]$ of a 6-line four-loop vacuum integral in D = 2-2δ
PR9D2Eps1	$[\delta^1]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR9 in D = 2−2δ, unit masses, Euclidean	**5531**	no	68 (4L 45, 5L 23)	$[\delta^1]$ of a 7-line four-loop vacuum integral in D = 2-2δ
PR9D2Eps2	$[\delta^2]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR9 in D = 2−2δ, unit masses, Euclidean	**5529**	no	53 (4L 39, 5L 14)	$[\delta^2]$ of a 7-line four-loop vacuum integral in D = 2-2δ
PR9D2Eps3	$[\delta^3]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR9 in D = 2−2δ, unit masses, Euclidean	**5527**	no	25 (4L 19, 5L 6)	$[\delta^3]$ of a 7-line four-loop vacuum integral in D = 2-2δ
PR9pD2Eps2	$[\delta^2]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR9p (one dotted line) in D = 2−2δ, unit masses, Euclidean	**5527**	no	51 (4L 37, 5L 14)	$[\delta^2]$ of a 7-line four-loop vacuum integral in D = 2-2δ
PR9D2Eps4	$[\delta^4]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR9 in D = 2−2δ, unit masses, Euclidean	**5525**	no	4 (4L 3, 5L 1)	$[\delta^4]$ of a 7-line four-loop vacuum integral in D = 2-2δ
PR9pD2Eps3	$[\delta^3]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR9p (one dotted line) in D = 2−2δ, unit masses, Euclidean	**5525**	no	25 (4L 19, 5L 6)	$[\delta^3]$ of a 7-line four-loop vacuum integral in D = 2-2δ
PR9pD2Eps4	$[\delta^4]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR9p (one dotted line) in D = 2−2δ, unit masses, Euclidean	**5524**	no	4 (4L 3, 5L 1)	$[\delta^4]$ of a 7-line four-loop vacuum integral in D = 2-2δ
PR10D2Eps1	$[\delta^1]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR10 in D = 2−2δ, unit masses, Euclidean	**5406**	no	48 (4L 28, 5L 20)	$[\delta^1]$ of a 7-line four-loop vacuum integral in D = 2-2δ
PR10D2Eps2	$[\delta^2]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR10 in D = 2−2δ, unit masses, Euclidean	**5406**	no	25 (4L 16, 5L 9)	$[\delta^2]$ of a 7-line four-loop vacuum integral in D = 2-2δ
PR10D2Eps3	$[\delta^3]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR10 in D = 2−2δ, unit masses, Euclidean	**5404**	no	13 (4L 10, 5L 3)	$[\delta^3]$ of a 7-line four-loop vacuum integral in D = 2-2δ
PR10D2Eps4	$[\delta^4]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR10 in D = 2−2δ, unit masses, Euclidean	**5402**	no	4 (4L 3, 5L 1)	$[\delta^4]$ of a 7-line four-loop vacuum integral in D = 2-2δ
PR11dD2Eps1	$[\delta^1]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR11d (one dotted line) in D = 2−2δ, unit masses, Euclidean	**5363**	no	37 (4L 20, 5L 17)	$[\delta^1]$ of an 8-line four-loop vacuum integral in D = 2-2δ
PR11D2Eps2	$[\delta^2]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR11 in D = 2−2δ, unit masses, Euclidean	**5361**	no	19 (4L 11, 5L 8)	$[\delta^2]$ of an 8-line four-loop vacuum integral in D = 2-2δ
PR11D2Eps3	$[\delta^3]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR11 in D = 2−2δ, unit masses, Euclidean	**5359**	no	13 (4L 10, 5L 3)	$[\delta^3]$ of an 8-line four-loop vacuum integral in D = 2-2δ
PR11D2Eps4	$[\delta^4]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR11 in D = 2−2δ, unit masses, Euclidean	**5359**	no	4 (4L 3, 5L 1)	$[\delta^4]$ of an 8-line four-loop vacuum integral in D = 2-2δ
PR11dD2Eps3	$[\delta^3]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR11d (one dotted line) in D = 2−2δ, unit masses, Euclidean	**5358**	no	13 (4L 10, 5L 3)	$[\delta^3]$ of an 8-line four-loop vacuum integral in D = 2-2δ
PR11dD2Eps4	$[\delta^4]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR11d (one dotted line) in D = 2−2δ, unit masses, Euclidean	**5357**	no	4 (4L 3, 5L 1)	$[\delta^4]$ of an 8-line four-loop vacuum integral in D = 2-2δ
PR15D2Eps1	$[\delta^1]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR15 in D = 2−2δ, unit masses, Euclidean	**2805**	no	6 (4L 4, 5L 2)	$[\delta^1]$ of an 8-line four-loop vacuum integral in D = 2-2δ
PR15D2Eps2	$[\delta^2]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR15 in D = 2−2δ, unit masses, Euclidean	**2601**	no	9 (4L 4, 5L 5)	$[\delta^2]$ of an 8-line four-loop vacuum integral in D = 2-2δ
PR12D2Eps2	$[\delta^2]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR12 in D = 2−2δ, unit masses, Euclidean	**2600**	no	6 (4L 2, 5L 4)	$[\delta^2]$ of a 9-line four-loop vacuum integral in D = 2-2δ
PR0D2Eps2	$[\delta^2]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR0 in D = 2−2δ, unit masses, Euclidean	**2396**	no	2 (4L 1, 5L 1)	$[\delta^2]$ of a 9-line four-loop vacuum integral in D = 2-2δ
PR0D2Eps1	$[\delta^1]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR0 in D = 2−2δ, unit masses, Euclidean	**2395**	no	5 (4L 2, 5L 3)	$[\delta^1]$ of a 9-line four-loop vacuum integral in D = 2-2δ
PR15D2Eps3	$[\delta^3]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR15 in D = 2−2δ, unit masses, Euclidean	**2351**	no	2 (4L 1, 5L 1)	$[\delta^3]$ of an 8-line four-loop vacuum integral in D = 2-2δ
PR12D2Eps3	$[\delta^3]\ e^{4\gamma_E \delta}$ × FMFT four-loop topology PR12 in D = 2−2δ, unit masses, Euclidean	**2348**	no	2 (4L 1, 5L 1)	$[\delta^3]$ of a 9-line four-loop vacuum integral in D = 2-2δ

Appendix: constants, traditional forms and definitions — verified to the full stored digits; not yet adopted in the data. Formulas with more than 20 terms (D5bar2, D6bar2) are in `docs/benchmark/data/CONSTANTS-TRADITIONAL.md`. Notation: w = exp(iπ/3); Li_{a1..ar}(x1..xr) = Σ_{k1>..>kr≥1} Π_i x_i^k_i / k_i^a_i; Ls_j^(k)(t) = −∫_0^t φ^k ln^(j−1−k) |2 sin(φ/2)| dφ. Loop integrals (Euclidean, unit masses; every propagator 1/(q_e^2 + 1)^{a_e}; measure prod_l d^D k_l / pi^{D/2}; X_hat(delta) = e^{4 gamma_E delta} x integral in D = 2 - 2 delta; PRnD2EpsJ = [delta^J] X_hat; PRnD2 = X_hat(0); PR14epJ = [eps^J] of (-1)^{sum a} e^{4 gamma_E eps} x integral in D = 4 - 2 eps):

cl2	cl2 = Cl2(π/3) (agrees to 20004.3 d with the stored 20000 d)
cl4	cl4 = Cl4(π/3) (agrees to 20005.8 d with the stored 20000 d)
D5bar1	D5bar1 = - 4/3 ln2^4 ln3 + 4/3 ln2^2 ln3 π^2 - 14 ln2^2 ζ(3) - 1/15 ln3^5 + 2/9 ln3^3 π^2 + 4/15 ln3^2 π Cl2(π/3) + 29/45 ln3 π^4 - 32 ln3 Li4(1/2) + 68/135 π^3 Cl2(π/3) + 7/54 π^2 ζ(3) + 16/3 π Cl4(π/3) - 28 Li2(-1/2) ζ(3) - 64/5 Cl2(π/3) Im Li3(i/√3) - 2798/81 ζ(5) + 16 Li5(1/3) - 8 Li5(-1/3) - 32 Re -Li_{3,1,1}(-1,1,w) (agrees to 20008.7 d with the stored 19998 d)
D5bar2	111 terms: see CONSTANTS-TRADITIONAL.md (agrees to 20008.0 d with the stored 19998 d)
D6bar1	34 terms: see CONSTANTS-TRADITIONAL.md (agrees to 20008.4 d with the stored 19998 d)
D6bar2	118 terms: see CONSTANTS-TRADITIONAL.md (agrees to 20008.0 d with the stored 19998 d)
s6	s6 = 1/208 ln2^6 - 1/208 ln2^4 π^2 + 1/156 ln2^2 π^4 + 89/58968 π^6 + 3/8 ζ(3)^2 + 126/13 Li6(1/2) + 162/13 Li6(-1/2) - 1/39 Li6(-1/8) (agrees to 20006.5 d with the stored 20000 d)
T111bar4	T111bar4 = 1/20 ln3^5 π - 3/4 ln3^4 Cl2(π/3) - 17/180 ln3^3 π^3 - 24/5 ln3^3 Im Li3(i/√3) + 42/5 ln3^2 Cl4(π/3) - 72/5 ln3^2 Im Li4(i/√3) + 19/120 ln3 π^5 + 3 ln3 Ls5^0(2π/3) + π^3 ζ(3) + 18 π ζ(5) - 6/5 Ls6^0(2π/3) (agrees to 20003.9 d with the stored 19998 d)
K5LogMoment(1)	3 int_0^{inf} t^1 K_0(t)^5 (ln t + gamma_E - ln 2) dt = 3 (d/ds c_(5,s))|_(s=1) + 3 (gamma_E - ln 2) c_(5,1)
K5LogMoment(3)	3 int_0^{inf} t^3 K_0(t)^5 (ln t + gamma_E - ln 2) dt = 3 (d/ds c_(5,s))|_(s=3) + 3 (gamma_E - ln 2) c_(5,3)
K5Moment(1)	c_(5,1) = int_0^{inf} t^1 K_0(t)^5 dt (Bessel moment, Broadhurst notation)
K5Moment(3)	c_(5,3) = int_0^{inf} t^3 K_0(t)^5 dt (Bessel moment, Broadhurst notation)
T111bar5	T111bar5 = - 1/48 ln3^6 π + 3/10 ln3^5 Cl2(π/3) + 17/720 ln3^4 π^3 + 9/5 ln3^4 Im Li3(i/√3) - 14/5 ln3^3 Cl4(π/3) + 24/5 ln3^3 Im Li4(i/√3) - 19/240 ln3^2 π^5 - 3/2 ln3^2 Ls5^0(2π/3) - ln3 π^3 ζ(3) - 18 ln3 π ζ(5) + 6/5 ln3 Ls6^0(2π/3) - 55/1008 π^7 - 6 π ζ(3)^2 - 2/5 Ls7^0(2π/3) (agrees to 20003.4 d with the stored 19998 d) Definition: T111bar(5) in the MATAD convention of vakint matad.rs: sum_k T111bar(k) eps^k = sqrt3 (1-2eps)(1-eps) Sunset2(eps)/G(eps)^2 - 3 sqrt3/(2 eps^2), G = exp(EulerGamma eps) Gamma(1+eps), Sunset2 = -miT111, miT111 = exp(2 EulerGamma eps) * integral d^{dk} d^{dl}/pi^d [(k^2+1) (l^2+1)((k+l)^2+1)]$^{-1}$ (Euclidean, d = 4-2eps); weight-7 coefficient at eps^5
BNbar5	BNbar5 = - 768/35 ln2^7 + 384/5 ln2^5 π^2 + 1088/5 ln2^3 π^4 + 44640 ln2^2 ζ(5) + 2944/105 ln2 π^6 + 19520 ln2 ζ(3)^2 - 46080 ln2 Re Li5,1(-1,-1) - 688/3 π^4 ζ(3) - 9440/3 π^2 ζ(5) - 47373 ζ(7) + 110592 Li7(1/2) - 40960 Re -Li_{5,1,1}(-1,1,1) - 21760 Re -Li_{4,2,1}(-1,1,1) (agrees to 19712.3 d with the stored 19706 d) Definition: BNbar(5) in the MATAD convention of vakint matad.rs: miBN = exp(3 EulerGamma eps) Gamma(1+eps)^3 (4/eps^3 - 44/(3 eps^2) + sum_(k>=1) BNbar(k) eps^k)/((1-3eps)(2-3eps)(1-2eps)(1-eps)), miBN = exp(3 EulerGamma eps) * integral of the three-loop equal-mass banana (four unit-mass lines k1, k2, k3, k1+k2+k3; Euclidean, d^{dk}/pi$^{d/2}$ per loop); weight-7 coefficient at eps^5
K5MomentTaylor(1,2)	[eps^2] e$^{3 (gamma_E - ln 2} eps) int_0^{inf} t^{1+3eps} K_eps(t)^5 dt (ordinary Taylor coefficient; [eps^0] = K5Moment(1), [eps^1] = K5LogMoment(1))
K5MomentTaylor(1,3)	[eps^3] e$^{3 (gamma_E - ln 2} eps) int_0^{inf} t^{1+3eps} K_eps(t)^5 dt (ordinary Taylor coefficient; [eps^0] = K5Moment(1), [eps^1] = K5LogMoment(1))
K5MomentTaylor(1,4)	[eps^4] e$^{3 (gamma_E - ln 2} eps) int_0^{inf} t^{1+3eps} K_eps(t)^5 dt (ordinary Taylor coefficient; [eps^0] = K5Moment(1), [eps^1] = K5LogMoment(1))
K5MomentTaylor(3,2)	[eps^2] e$^{3 (gamma_E - ln 2} eps) int_0^{inf} t^{3+3eps} K_eps(t)^5 dt (ordinary Taylor coefficient; [eps^0] = K5Moment(3), [eps^1] = K5LogMoment(3))
K5MomentTaylor(3,3)	[eps^3] e$^{3 (gamma_E - ln 2} eps) int_0^{inf} t^{3+3eps} K_eps(t)^5 dt (ordinary Taylor coefficient; [eps^0] = K5Moment(3), [eps^1] = K5LogMoment(3))
K5MomentTaylor(3,4)	[eps^4] e$^{3 (gamma_E - ln 2} eps) int_0^{inf} t^{3+3eps} K_eps(t)^5 dt (ordinary Taylor coefficient; [eps^0] = K5Moment(3), [eps^1] = K5LogMoment(3))
PR14ep2	[ep^2] D=4-2eps; 1/([(k1 -k2)^2+1] [(k1 -k3 +k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3)^2+1] [(k1)^2+1] [(k3)^2+1] [(k4)^2+1]) (7 propagators)
D5bar3 **undecided**	D5bar(3) in the MATAD convention of vakint matad.rs: miD5 = (D5bar(0) + eps D5bar(1) + eps^2 D5bar(2) + eps^3 D5bar(3) + ... + 2 zeta3/eps)/((1-2eps)(1-eps)), miD5 = exp(3 EulerGamma eps) * integral of the three-loop tetrahedron with five unit-mass lines and one massless line (Euclidean, d^{dk}/pi$^{d/2}$ per loop, d = 4-2eps); weight-7 coefficient at eps^3
PR14ep3	[ep^3] D=4-2eps; 1/([(k1 -k2)^2+1] [(k1 -k3 +k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3)^2+1] [(k1)^2+1] [(k3)^2+1] [(k4)^2+1]) (7 propagators)
PR5D2Eps4	[delta^4] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1]) (6 propagators)
PR7D2	(delta^0 value) D=2; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1] [(k4)^2+1]) (6 propagators)
PR7D2Eps1	[delta^1] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1] [(k4)^2+1]) (6 propagators)
PR7D2Eps2	[delta^2] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1] [(k4)^2+1]) (6 propagators)
PR7D2Eps3	[delta^3] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1] [(k4)^2+1]) (6 propagators)
PR7D2Eps4	[delta^4] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1] [(k4)^2+1]) (6 propagators)
PR9D2Eps1	[delta^1] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1] [(k4)^2+1]) (7 propagators)
PR9D2Eps2	[delta^2] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1] [(k4)^2+1]) (7 propagators)
PR9D2Eps3	[delta^3] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1] [(k4)^2+1]) (7 propagators)
PR9pD2Eps2	[delta^2] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1]^2 [(k4)^2+1]) (7 propagators)
PR9D2Eps4	[delta^4] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1] [(k4)^2+1]) (7 propagators)
PR9pD2Eps3	[delta^3] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1]^2 [(k4)^2+1]) (7 propagators)
PR9pD2Eps4	[delta^4] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3)^2+1]^2 [(k4)^2+1]) (7 propagators)
PR10D2Eps1	[delta^1] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k2)^2+1] [(k4)^2+1]) (7 propagators)
PR10D2Eps2	[delta^2] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k2)^2+1] [(k4)^2+1]) (7 propagators)
PR10D2Eps3	[delta^3] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k2)^2+1] [(k4)^2+1]) (7 propagators)
PR10D2Eps4	[delta^4] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k2)^2+1] [(k4)^2+1]) (7 propagators)
PR11dD2Eps1	[delta^1] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k2)^2+1]^2 [(k3)^2+1] [(k4)^2+1]) (8 propagators)
PR11D2Eps2	[delta^2] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k2)^2+1] [(k3)^2+1] [(k4)^2+1]) (8 propagators)
PR11D2Eps3	[delta^3] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k2)^2+1] [(k3)^2+1] [(k4)^2+1]) (8 propagators)
PR11D2Eps4	[delta^4] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k2)^2+1] [(k3)^2+1] [(k4)^2+1]) (8 propagators)
PR11dD2Eps3	[delta^3] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k2)^2+1]^2 [(k3)^2+1] [(k4)^2+1]) (8 propagators)
PR11dD2Eps4	[delta^4] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k2)^2+1]^2 [(k3)^2+1] [(k4)^2+1]) (8 propagators)
PR15D2Eps1	[delta^1] D=2-2delta; 1/([(k1 -k2)^2+1] [(k1 -k3 +k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3)^2+1] [(k1)^2+1] [(k2)^2+1] [(k3)^2+1] [(k4)^2+1]) (8 propagators)
PR15D2Eps2	[delta^2] D=2-2delta; 1/([(k1 -k2)^2+1] [(k1 -k3 +k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3)^2+1] [(k1)^2+1] [(k2)^2+1] [(k3)^2+1] [(k4)^2+1]) (8 propagators)
PR12D2Eps2	[delta^2] D=2-2delta; 1/([(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3 +k4)^2+1] [(k1)^2+1] [(k2)^2+1] [(k3)^2+1] [(k4)^2+1]) (9 propagators)
PR0D2Eps2	[delta^2] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k1)^2+1] [(k2)^2+1] [(k3)^2+1] [(k4)^2+1]) (9 propagators)
PR0D2Eps1	[delta^1] D=2-2delta; 1/([(k1 +k2 -k3 -k4)^2+1] [(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k1)^2+1] [(k2)^2+1] [(k3)^2+1] [(k4)^2+1]) (9 propagators)
PR15D2Eps3	[delta^3] D=2-2delta; 1/([(k1 -k2)^2+1] [(k1 -k3 +k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3)^2+1] [(k1)^2+1] [(k2)^2+1] [(k3)^2+1] [(k4)^2+1]) (8 propagators)
PR12D2Eps3	[delta^3] D=2-2delta; 1/([(k1 -k3 -k4)^2+1] [(k1 -k3)^2+1] [(k2 -k3 -k4)^2+1] [(k2 -k3)^2+1] [(k3 +k4)^2+1] [(k1)^2+1] [(k2)^2+1] [(k3)^2+1] [(k4)^2+1]) (9 propagators)

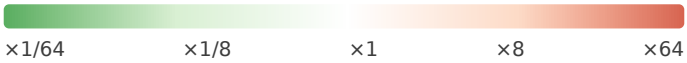
Reading guide. Bold: RustRed median, then the reference's (M = MATAD, F = FMFT, αL = alphaLoop) and the ratio RustRed ÷ reference; green = RustRed faster, red = slower (log scale); (·) = minimum over the runs. *Reduction*: masters retained; *full*: complete call with master values; *warm*: the identical call repeated in the process (step cache filled). Every timed call follows an untimed warm-up on another integral, so one-time setup is excluded (listed below the table). *IBP*: RustRed's time in the reduction itself. Five loops (no FORM backend): a master is recognised in any form and returned without a reduction walk; other inputs run RustRed's modular ordered walk (`modular_sampled`, on the cores a cell names).

Timings

Table 1: **RustRed versus the FORM backends** (MATAD and alphaLoop: one to three loops; FMFT: four loops; none reaches five). Terminals: distinct RustRed master symbols in the retained result (one to three loops: terminals of the closed program, symmetric copies included). Agreement: digits shared with RustRed at $m = \mu = 1$, worst order of ε.

Test case	Terminals	Reduction vs M / F	Reduction vs αL	Warm reduction	Full call vs M / F	Full call vs αL	Agreement
One loop							
QCD ladder, q = 0 quark loop, no rung	1	**4.35 ms** (3.24) M 24.1 ms (23.2) **×0.18** IBP 1.51 ms	**4.35 ms** (3.24) αL 22.5 ms (22.0) **×0.19**	**4.44 ms** (3.19) cold 4.35 ms **×1.0**	**23.3 ms** (18.6) M 160 ms (158) **×0.15**	**23.3 ms** (18.6) αL 39.6 ms (39.3) **×0.59**	**31** digits
QCD ladder, O(q²) quark loop, no rung	1	**5.14 ms** (4.75) M 43.6 ms (41.4) **×0.12** IBP 1.61 ms	**5.14 ms** (4.75) αL 39.7 ms (36.8) **×0.13**	**4.80 ms** (3.90) cold 5.14 ms **×0.93**	**21.7 ms** (18.2) M 301 ms (300) **×1/14**	**21.7 ms** (18.2) αL 71.3 ms (68.6) **×0.30**	**31** digits
Tadpole T1 power 5 · rank 4	1	**4.26 ms** (4.26) M 22.3 ms (22.2) **×0.19** IBP 1.81 ms	**4.26 ms** (4.26) αL 22.9 ms (22.8) **×0.19**	**4.33 ms** (3.40) cold 4.26 ms **×1.0**	**12.2 ms** (11.0) M 154 ms (153) **×1/13**	**12.2 ms** (11.0) αL 32.1 ms (31.4) **×0.38**	**31** digits
Two loops							
QCD ladder, q = 0 1 gluon rung	2	**22.0 ms** (21.5) M 35.9 ms (34.1) **×0.61** IBP 5.87 ms	**22.0 ms** (21.5) αL 29.9 ms (29.9) **×0.74**	**18.9 ms** (18.4) cold 22.0 ms **×0.86**	**118 ms** (110) M 211 ms (204) **×0.56**	**118 ms** (110) αL 65.8 ms (65.0) **×1.8**	**31** digits
QCD ladder, O(q²) 1 gluon rung	2	**146 ms** (132) M 281 ms (278) **×0.52** IBP 9.65 ms	**146 ms** (132) αL 225 ms (197) **×0.65**	**139 ms** (134) cold 146 ms **×0.96**	**243 ms** (207) M 1.12 s (1.11) **×0.22**	**243 ms** (207) αL 299 ms (292) **×0.81**	**31** digits
Sunset T111 powers 3,2,2 · rank 4	2	**5.15 ms** (4.88) M 33.6 ms (30.8) **×0.15** IBP 1.30 ms	**5.15 ms** (4.88) αL 26.1 ms (24.5) **×0.20**	**5.18 ms** (4.89) cold 5.15 ms **×1.0**	**20.0 ms** (19.7) M 172 ms (172) **×0.12**	**20.0 ms** (19.7) αL 35.4 ms (33.2) **×0.57**	**31** digits
Three loops							
QCD ladder, q = 0 2 gluon rungs	14	**87.7 ms** (84.2) M 182 ms (181) **×0.48** IBP 65.5 ms	**87.7 ms** (84.2) αL 162 ms (162) **×0.54**	**49.4 ms** (49.3) cold 87.7 ms **×0.56**	**1.81 s** (1.80) M 1.10 s (1.08) **×1.6**	**1.81 s** (1.80) αL 277 ms (276) **×6.5**	**31** digits
QCD ladder, O(q²) 2 gluon rungs	14	**679 ms** (644) M 4.26 s (4.24) **×0.16** IBP 257 ms	**679 ms** (644) αL 4.71 s (4.69) **×0.14**	**486 ms** (483) cold 679 ms **×0.72**	**2.77 s** (2.65) M 11.9 s (11.7) **×0.23**	**2.77 s** (2.65) αL 5.01 s (4.85) **×0.55**	**31** digits
Tetrahedron D6 master	1	**12.0 ms** (11.8) M 68.4 ms (66.9) **×0.18** IBP 1.03 ms	**12.0 ms** (11.8) αL 26.4 ms (26.3) **×0.45**	**7.40 ms** (7.37) cold 12.0 ms **×0.62**	**25.9 ms** (25.2) M 252 ms (251) **×0.10**	**25.9 ms** (25.2) αL 59.5 ms (55.3) **×0.44**	**31** digits
Tetrahedron D6 3 dots · rank 4	38	**128 ms** (122) M 82.4 ms (82.3) **×1.6** IBP 118 ms	**128 ms** (122) αL 137 ms (136) **×0.93**	**72.7 ms** (71.3) cold 128 ms **×0.57**	**5.67 s** (5.52) M failed	**5.67 s** (5.52) αL 194 ms (194) **×29**	**31** digits
Four loops							
QCD ladder, q = 0 3 gluon rungs	10	**1.10 s** (1.02) F 5.84 s (5.82) **×0.19** IBP 336 ms	n/a	**924 ms** (864) cold 1.10 s **×0.84**	**7.44 s** (7.00) F 6.07 s (5.94) **×1.2**	n/a	**26** digits
QCD ladder, O(q²) 3 gluon rungs	10	**53.2 s** (52.0) F 467 s (457) **×0.11** IBP 1.50 s	n/a	**54.5 s** (54.3) cold 53.2 s **×1.0**	**61.9 s** (61.1) F 465 s (465) **×0.13**	n/a	**24** digits
H family one line squared	15	**927 ms** (896) F 609 ms (602) **×1.5** IBP 853 ms	n/a	**433 ms** (413) cold 927 ms **×0.47**	**1.29 s** (1.26) F 632 ms (628) **×2.0**	n/a	**22** digits
FG family one line to the 4th	15	**225 ms** (224) F 366 ms (365) **×0.61** IBP 169 ms	n/a	**121 ms** (117) cold 225 ms **×0.54**	**2.70 s** (2.68) F 403 ms (399) **×6.7**	n/a	**23** digits

Test case	Termi-nals	Reduction vs M / F	Reduction vs αL	Warm reduction	Full call vs M / F	Full call vs αL	Agree-ment
X family one line to the 4th	19	**4.86 s** (4.82) F 17.8 s (17.6) **×0.27** IBP 4.66 s	n/a	**3.04 s** (2.99) cold 4.86 s **×0.63**	**5.73 s** (5.57) F 17.8 s (17.7) **×0.32**	n/a	**21** digits
Five loops							
QCD ladder, q = 0 4 gluon rungs	79	**60.2 s**† (55.9) no FORM backend on 8 cores	n/a	**34.0 s**† (32.5) cold 60.2 s† **×0.57** on 8 cores	**3.49 h**† 1 run no FORM backend on 8 cores	n/a	—
QCD ladder, O(q²) 4 gluon rungs	79	**15.2 min**† 1 run no FORM backend on 8 cores	n/a	**574 s**† 1 run cold 15.2 min† **×0.63** on 8 cores	**6.16 h**† 1 run no FORM backend on 8 cores	n/a	—
Top-sector master (P105) 12 lines · scalar	1	**151 ms** (143) no FORM backend IBP 642 µs	n/a	**156 ms** (144) cold 151 ms **×1.0**	**154 ms** (144) no FORM backend	n/a	—
Dotted master (P63) 10 lines · rank 1	1	**144 ms** (137) no FORM backend IBP 599 µs	n/a	**130 ms** (127) cold 144 ms **×0.90**	**347 ms** (326) no FORM backend	n/a	—
Top sector (P105) master + one dot	98	**503 s**† 1 run no FORM backend on 8 cores	n/a	**300 s** 1 run cold 503 s† **×0.60** on 8 cores	failed no FORM backend	n/a	—
Dotted-master sector (P63) master + one dot	79	**54.8 s** (54.2) no FORM backend on 8 cores	n/a	**25.8 s** (25.5) cold 54.8 s **×0.47** on 8 cores	failed no FORM backend	n/a	—



† **Median not timed in an idle window** (the other cells' medians were, some after re-measurement): 5L QCD ladder, q = 0 · reduction: other tasks used 29.8% of each of the 8 timed CPUs; 5L QCD ladder, q = 0 · warm reduction: other tasks used 14.2% of each of the 8 timed CPUs; 5L QCD ladder, q = 0 · full call: one 3.5 h run (2026-11-06 22:44Z to 11-07 02:14Z) alongside other benchmark lanes of ours on other CPUs; other tasks used 21.0% of each of the 8 timed CPUs; 5L QCD ladder, O(q²) · reduction: other tasks used 4.0% of each of the 8 timed CPUs; 5L QCD ladder, O(q²) · warm reduction: other tasks used 4.0% of each of the 8 timed CPUs; 5L QCD ladder, O(q²) · full call: one 6.2 h run (2026-11-07 02:14Z to 08:25Z) alongside other benchmark and integration lanes of ours on other CPUs; 5L Top sector · reduction: other tasks used 7.7% of each of the 8 timed CPUs.

Once per process (excluded from every cell; first-call setup = first minus second call on the warm-up integral)

	1L	2L	3L	4L	5L
RustRed data load	1.85 s	1.82 s	869 ms	8.05 s	39.0 s
RustRed full call	1.93 ms	2.63 ms	22.6 ms	39.7 ms	862 ms
MATAD full call	928 ms	941 ms	870 ms	—	—
alphaLoop full call	892 ms	897 ms	809 ms	—	—
FMFT full call	—	—	—	813 ms	—

Failed calls.
3L Tetrahedron D6, 3 dots · rank 4 · MATAD full call (3/3 runs): vakint's precision guard rejects the backend's result: the imaginary part of the coefficient of 1/(mUVsq*ε^2) is −1.9e−71 ± 1.2e−70 (consistent with zero, no reliable digit)
5L Top sector (P105), master + one dot · full call (1/1 runs): vakint fails closed (MissingMasterOrders): 72 masters lack Laurent orders this result needs (the reduction coefficients have poles at d = 4; deepest orders needed: 5L to ε^1, 4L to ε^4, 3L to ε^3, 1L to ε^6), so the result fails from ε^−2 on
5L Dotted-master sector (P63), master + one dot · full call (1/1 runs): vakint's precision guard rejects the backend's result: the imaginary part of the coefficient of 1/(mUV^2*ε^3) is −2.2e−71 ± 3.0e−69 (consistent with zero, no reliable digit)

Provenance

Data (timings)	vakint-external-data `v0.2.1` · commit `fb978fb9aa55` · root manifest `87715e98cd999ca0`
Data (tables)	vakint-external-data v0.2.2 (0faa1c795664): precision tables regenerated from this later release, whose reduction payloads equal the timed tree's (receipt rule 2)
Vakint	0.2.0 · release build of `71316be4ff16` (clean) before the rebase onto the data pin; same source as `4852765c8259` on this branch outside docs and vendored data (receipt rule 1) · harness `951c0005eac19098` · FORM 5.0.0
Host	AMD EPYC 9754 128-Core Processor, 384 logical CPUs, shared; timed on 42 pinned CPUs in 9 L3 domains (listed in suite.json)
Contention	318 samples, 12 still contended after re-measurement († thresholds per timed CPU: SMT sibling 25% busy, other tasks 3%, call run-queue wait 3% when other tasks ran) · medians (max): SMT sibling busy 2.4% (23.3%), timed call in the run queue 0.2% (11%), other tasks per timed CPU 1.2% (47.3%) · 1-min load ≤ 294.6
Regenerate	`crates/vakint/docs/benchmark/make-benchmark.sh DATA_ROOT BINARY OUT --install` after a data release `DATA_ROOT - OUT --refresh-tables --install` (timings kept) · assembled 2026-11-07T11:37:06Z

Reduction graphs

Nodes are the vacuum topologies of the shipped reduction, on the row of their number of lines; an edge removes one line (thicker: several lines), as in Fig. 21 of B. Ruijl's thesis (Leiden, 2017). Products (one-vertex-reducible graphs) are merged by their factors; numbers count masters (one to three loops: RustRed terminals).

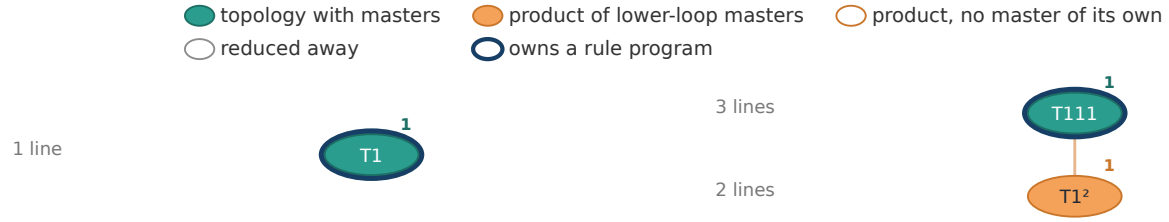


Figure 1: **1-loop reduction graph.** T1 carries the one-loop master; dots and numerators reduce inside it. 1 topology · 0 edges · 1 terminal.

Figure 2: **2-loop reduction graph.** Removing any sunset line gives T1². 2 topologies · 1 edge · 2 terminals.

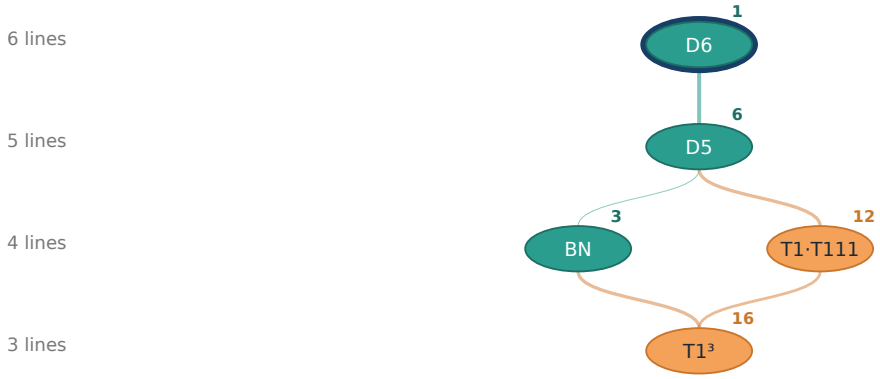


Figure 3: **3-loop reduction graph.** MATAD names; symmetric sectors are separate terminals of the closed program. 5 topologies · 5 edges · 38 terminals.

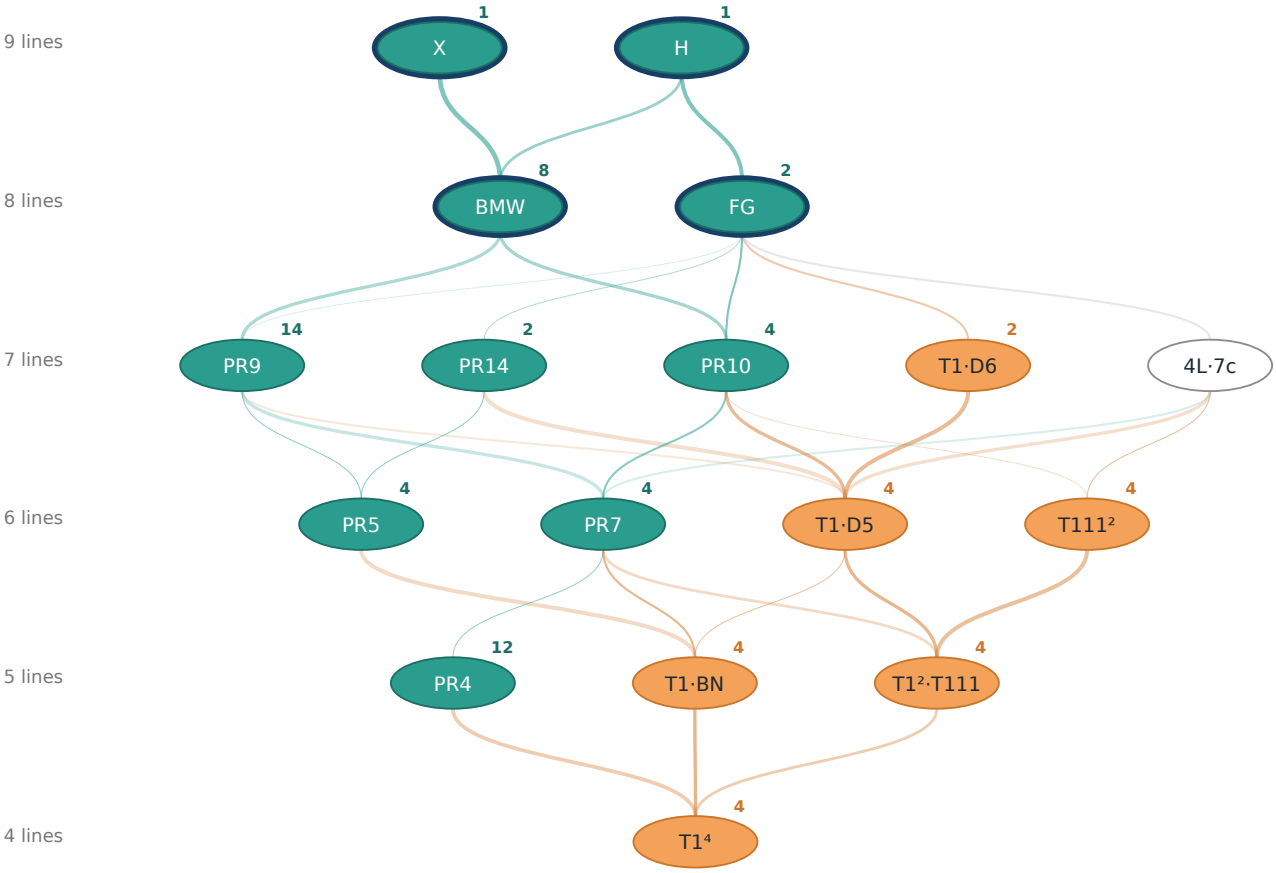


Figure 4: **4-loop reduction graph.** Rings: the FMFT family programs H, X, BMW, FG; irreducible topologies named by their FMFT master PRk. 17 topologies · 32 edges · 74 masters.

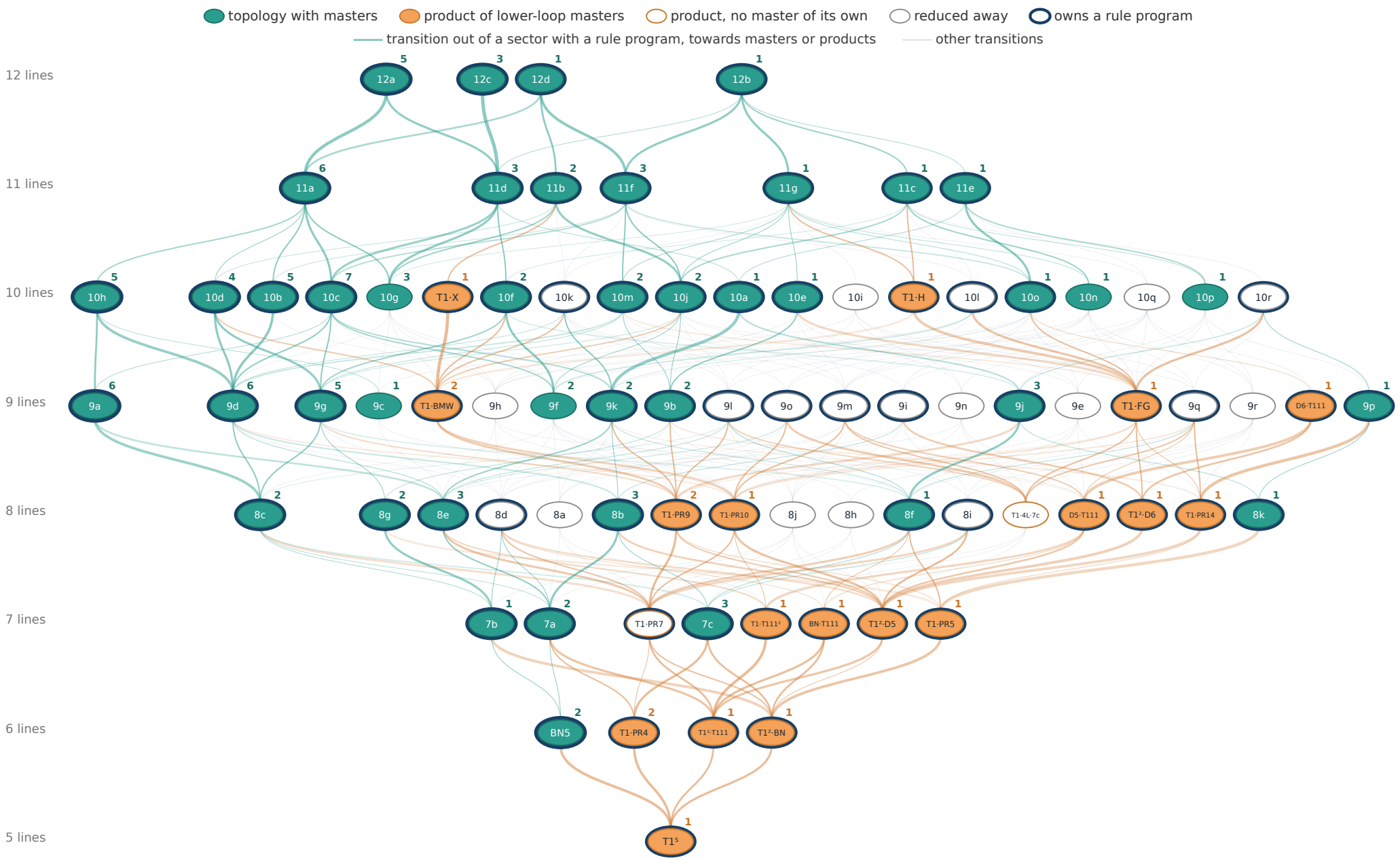


Figure 5: **5-loop reduction graph.** Rings: the 67 owner sectors of the saved application (width grows with their rule tables); names: lines and a letter. Benchmark masters: P105 in 12b, P63 in 10c. 82 topologies · 311 edges · 131 masters.

Precision of the masters and of the results

FINAL · data tree vakint-external-data v0.2.2 (0faa1c795664) (git 0faa1c795664, tree b738993f1fe0) · v0.2.2 root manifest 5ce0e4a0abe8687c · 4L constants 1338cfe501ffb516 · 5L values build 2026-11-07T02:24:15Z (producer-update 781d858170cd9127) · generated 2026-11-11T03:37:29Z · tables 7c9148d895cb5b0e

Class of every Laurent coefficient, worst case X rounded down. **Precision table 1**: every master the data tree ships, at every order it holds. **Precision table 2**: the result of any single-scale vacuum input with L loops (RustRed backend), worst case over the inputs; an order that needs master coefficients the data do not hold (spurious poles of the reduction, lower-loop factors of 5L products) is not available.

S exact, periods at any precision · **0** structural zero (table 1) · **S*(X)** exact, periods to at most X digits · **N(X)** numerical only, X = validated digits − 1 · **_** not available

Precision table 1. Master coefficients — every master of the data tree (rows, by loop count; 5L: the 640-pack name in brackets) at every Laurent order held (columns); — = not held; 0 = structural zero (exact zero row). [p] X rests in part on a constant computed with two profiles of one code beyond its independent confirmation; [n] Numerical mode returns more digits.

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
1L · 1 master · 7 S, 0 zero, 0 S*, 0 N											
1L/P1	—	—	—	—	S	S	S	S	S	S	S
2L · 2 masters · 13 S, 0 zero, 0 S*, 0 N											
2L/P1	—	—	—	S	S	S	S	S	S	—	—
2L/P2	—	—	—	S	S	S	S	S	S	S	—
3L · 38 masters · 250 S, 2 zero, 0 S*, 0 N											
3L/P1	—	—	S	S	S	S	S	S	—	—	—
3L/P2	—	—	S	S	S	S	S	S	—	—	—
3L/P3	—	—	S	S	S	S	S	S	—	—	—
3L/P4	—	—	S	S	S	S	S	S	S	—	—
3L/P5	—	—	S	S	S	S	S	S	—	—	—
3L/P6	—	—	S	S	S	S	S	S	—	—	—
3L/P7	—	—	S	S	S	S	S	S	—	—	—
3L/P8	—	—	S	S	S	S	S	S	S	—	—
3L/P9	—	—	S	S	S	S	S	S	—	—	—
3L/P10	—	—	S	S	S	S	S	S	S	—	—
3L/P11	—	—	S	S	S	S	S	S	—	—	—
3L/P12	—	—	S	S	S	S	S	S	S	—	—
3L/P13	—	—	S	S	S	S	S	S	S	S	—
3L/P14	—	—	S	S	S	S	S	S	S	—	—
3L/P15	—	—	S	S	S	S	S	S	—	—	—
3L/P16	—	—	S	S	S	S	S	S	—	—	—
3L/P17	—	—	S	S	S	S	S	S	—	—	—
3L/P18	—	—	S	S	S	S	S	S	S	—	—
3L/P19	—	—	S	S	S	S	S	S	—	—	—
3L/P20	—	—	S	S	S	S	S	S	—	—	—
3L/P21	—	—	S	S	S	S	S	S	S	S	—
3L/P22	—	—	S	S	S	S	S	S	S	—	—
3L/P23	—	—	S	S	S	S	S	S	S	—	—
3L/P24	—	—	S	S	S	S	S	S	S	—	—
3L/P25	—	—	S	S	S	S	S	S	—	—	—
3L/P26	—	—	S	S	S	S	S	S	S	—	—
3L/P27	—	—	S	S	S	S	S	S	—	—	—
3L/P28	—	—	S	S	S	S	S	S	S	S	—
3L/P29	—	—	S	S	S	S	S	S	S	—	—
3L/P30	—	—	S	S	S	S	S	S	S	—	—
3L/P31	—	—	S	S	S	S	S	S	—	—	—
3L/P32	—	—	S	S	S	S	S	S	S	—	—
3L/P33	—	—	S	S	S	S	S	S	S	—	—
3L/P34	—	—	S	S	S	S	S	S	S	—	—
3L/P35	—	—	S	S	S	S	S	S	S	—	—
3L/P36	—	—	S	S	S	S	S	S	S	—	—
3L/P37	—	—	S	S	S	S	S	S	S	—	—
3L/P38	—	—	0	0	S	S	S	S	—	—	—
4L · 74 masters · 321 S, 55 zero, 92 S*, 0 N											
4L/P1	—	S	S	S	S	S	S	S	—	—	—
4L/P2	—	S	S	S	S	S	S	S	S	—	—
4L/P3	—	S	S	S	S	S	S	S	—	—	—
4L/P4	—	S	S	S	S	S	S	S	—	—	—
4L/P5	—	S	S	S	S	S	S	—	—	—	—
4L/P6	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P7	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—
4L/P8	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P9	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P10	—	0	0	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P11	—	0	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P12	—	S	S	S	S	S	S	S	—	—	—
4L/P13	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P14	—	0	0	0	S	S*(5358)	S*(5357)[p]	—	—	—	—
4L/P15	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P16	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P17	—	0	0	0	0	S*(5363)[n]	S*(5358)	—	—	—	—
4L/P18	—	0	0	0	0	S*(2395)[p]	S*(2395)[p]	—	—	—	—
4L/P19	—	0	0	0	0	S*(5363)[n]	S*(5361)	—	—	—	—
4L/P20	—	S	S	S	S	S	S	S	—	—	—
4L/P21	—	S	S	S	S	S	S	S	S	—	—
4L/P22	—	S	S	S	S	S	S	S	—	—	—
4L/P23	—	S	S	S	S	S	S	S	—	—	—
4L/P24	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P25	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
4L/P26	—	S	S	S	S	S	S	S	—	—	—
4L/P27	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P28	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P29	—	0	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P30	—	S	S	S	S	S	S	—	—	—	—
4L/P31	—	S	S	S	S	S	S	—	—	—	—
4L/P32	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P33	—	0	0	S	S	S	S	—	—	—	—
4L/P34	—	0	0	S	S	S*(2805)^{p}	S*(2601)^{p}	—	—	—	—
4L/P35	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P36	—	S	S	S	S	S	S	S	—	—	—
4L/P37	—	S	S	S	S	S	S	S	S	—	—
4L/P38	—	S	S	S	S	S	S	S	—	—	—
4L/P39	—	S	S	S	S	S	S	S	—	—	—
4L/P40	—	S	S	S	S	S	S	—	—	—	—
4L/P41	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P42	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—
4L/P43	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P44	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P45	—	0	0	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P46	—	0	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P47	—	S	S	S	S	S	S	S	—	—	—
4L/P48	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P49	—	0	0	0	S	S*(5358)	S*(5357)^{p}	—	—	—	—
4L/P50	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P51	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P52	—	0	0	0	0	S*(5363)^{n}	S*(5358)	—	—	—	—
4L/P53	—	S	S	S	S	S	S	S	—	—	—
4L/P54	—	S	S	S	S	S	S	S	S	—	—
4L/P55	—	S	S	S	S	S	S	S	—	—	—
4L/P56	—	S	S	S	S	S	S	S	—	—	—
4L/P57	—	S	S	S	S	S	S	—	—	—	—
4L/P58	—	S	S	S	S	S	S	—	—	—	—
4L/P59	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P60	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—
4L/P61	—	S	S	S	S	S	S	S	—	—	—
4L/P62	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P63	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P64	—	0	0	S	S	S	S	—	—	—	—
4L/P65	—	0	0	S	S	S*(2805)^{p}	S*(2601)^{p}	—	—	—	—
4L/P66	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P67	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P68	—	0	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P69	—	0	0	0	S	S*(5358)	S*(5357)^{p}	—	—	—	—
4L/P70	—	0	0	0	0	S*(2600)^{p}	S*(2348)^{p}	—	—	—	—
4L/P71	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P72	—	0	0	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P73	—	0	0	0	0	S*(5363)^{n}	S*(5361)	—	—	—	—
4L/P74	—	0	0	0	0	S*(5363)^{n}	S*(5358)	—	—	—	—
5L · 131 masters · 271 S, 291 zero, 29 S*, 195 N											
5L/P1 (P21)	S	S	S	N(66)	N(57)	N(58)	—	—	—	—	—
5L/P2 (P29)	S	S	S	N(82)	N(82)	N(74)	—	—	—	—	—
5L/P3 (P33)	S	S	S	N(68)	N(63)	N(60)	—	—	—	—	—
5L/P4 (P37)	S	S	S	N(64)	N(57)	N(55)	—	—	—	—	—
5L/P5 (P49)	0	S	S	N(64)	N(57)	N(56)	—	—	—	—	—
5L/P6 (P60)	S	S	N(64)	N(64)	N(56)	N(56)	—	—	—	—	—
5L/P7 (P65)	S	S	S	S	S	S	—	—	—	—	—
5L/P8 (P67)	S	S	S	S	S	N(141)	—	—	—	—	—
5L/P9 (P69)	0	S	S	S	S	N(141)	—	—	—	—	—
5L/P10 (P73)	S	S	S	N(58)	N(56)	N(55)	—	—	—	—	—
5L/P11 (P76)	S	S	S	S	S	S	—	—	—	—	—
5L/P12	S	S	S	S	S	S	—	—	—	—	—
5L/P13	S	S	S	S	S	S	—	—	—	—	—
5L/P14 (P95)	0	0	0	0	S	N(56)	—	—	—	—	—
5L/P15 (P100)	0	0	0	0	0	N(58)	—	—	—	—	—
5L/P16 (P101)	S	S	S	S	S	S	—	—	—	—	—
5L/P17 (P111)	S	S	S	N(58)	N(55)	N(55)	—	—	—	—	—
5L/P18 (P115)	S	S	S	N(64)	N(61)	N(57)	—	—	—	—	—
5L/P19 (P124)	S	S	S	S	N(56)	N(57)	—	—	—	—	—
5L/P20 (P126)	S	S	S	S	N(64)	N(60)	—	—	—	—	—
5L/P21 (P129)	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—	—
5L/P22 (P131)	S	S	S	N(64)	N(56)	N(55)	—	—	—	—	—
5L/P23 (P134)	S	S	S	S	N(83)	N(74)	—	—	—	—	—
5L/P24 (P140)	0	0	0	S	N(56)	N(56)	—	—	—	—	—
5L/P25 (P142)	0	0	0	0	S	N(59)	—	—	—	—	—
5L/P26 (P144)	0	0	0	0	S*(5363)	N(56)	—	—	—	—	—
5L/P27 (P158)	0	0	S	S	S	S	—	—	—	—	—
5L/P28 (P159)	S	S	S	S	S	S	—	—	—	—	—

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
5L/P29 (P169)	0	0	S	S	N(55)	N(55)	—	—	—	—	—
5L/P30 (P170)	0	0	0	0	S	N(46)	—	—	—	—	—
5L/P31 (P171)	0	0	0	0	0	N(47)	—	—	—	—	—
5L/P32 (P172)	0	0	0	0	0	N(46)	—	—	—	—	—
5L/P33 (P178)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P34 (P179)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P35 (P182)	0	0	S	S	S*(2805)^{p}	S*(2601)^{p}	—	—	—	—	—
5L/P36 (P202)	0	0	0	0	S	N(68)	—	—	—	—	—
5L/P37 (P204)	0	0	0	0	0	N(55)	—	—	—	—	—
5L/P38 (P238)	0	S	S	S	N(55)	N(54)	—	—	—	—	—
5L/P39 (P245)	0	0	0	0	S*(5363)	N(56)	—	—	—	—	—
5L/P40 (P263)	0	0	S	S	N(54)	N(55)	—	—	—	—	—
5L/P41 (P264)	0	0	0	0	0	N(42)	—	—	—	—	—
5L/P42 (P299)	0	0	0	0	0	N(57)	—	—	—	—	—
5L/P43 (P312)	0	S	S	S	N(56)	N(56)	—	—	—	—	—
5L/P44 (P314)	0	S	S	S	S	N(73)	—	—	—	—	—
5L/P45 (P315)	0	0	0	0	0	N(65)	—	—	—	—	—
5L/P46 (P321)	0	0	0	S	N(56)	N(57)	—	—	—	—	—
5L/P47 (P352)	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—	—
5L/P48 (P353)	0	0	S	S	S	S	—	—	—	—	—
5L/P49 (P356)	S	S	S	S	S	S	—	—	—	—	—
5L/P50 (P357)	0	0	S	N(46)	N(46)	N(46)	—	—	—	—	—
5L/P51 (P359)	0	S	S	S	S*(20000)	S*(20000)	—	—	—	—	—
5L/P52 (P361)	S	S	S	N(64)	N(56)	N(56)	—	—	—	—	—
5L/P53 (P367)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P54 (P368)	S	S	S	S	N(81)	N(75)	—	—	—	—	—
5L/P55 (P369)	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—	—
5L/P56 (P370)	0	0	S	S	N(81)	N(73)	—	—	—	—	—
5L/P57 (P375)	S	S	S	S	N(81)	N(75)	—	—	—	—	—
5L/P58 (P379)	0	0	0	S	S	N(73)	—	—	—	—	—
5L/P59 (P382)	0	0	0	S	S*(5358)	S*(5357)^{p}	—	—	—	—	—
5L/P60 (P383)	S	S	S	N(64)	N(56)	N(56)	—	—	—	—	—
5L/P61 (P384)	0	0	0	0	S	N(56)	—	—	—	—	—
5L/P62 (P385)	0	0	0	0	0	N(55)	—	—	—	—	—
5L/P63 (P387)	S	S	S	N(59)	N(55)	N(50)	—	—	—	—	—
5L/P64 (P389)	0	0	0	0	0	N(55)	—	—	—	—	—
5L/P65 (P391)	0	0	0	0	0	N(55)	—	—	—	—	—
5L/P66 (P398)	S	S	S	N(65)	N(56)	N(58)	—	—	—	—	—
5L/P67 (P399)	0	0	0	0	S	N(56)	—	—	—	—	—
5L/P68 (P400)	0	0	0	0	0	N(2)	—	—	—	—	—
5L/P69 (P413)	S	S	S	S	N(82)	N(73)	—	—	—	—	—
5L/P70 (P418)	S	S	S	S	N(81)	N(73)	—	—	—	—	—
5L/P71	S	S	S	S	N(56)	N(57)	—	—	—	—	—
5L/P72	S	S	S	N(63)	N(56)	N(56)	—	—	—	—	—
5L/P73 (P422)	S	S	S	N(81)	N(81)	N(73)	—	—	—	—	—
5L/P74 (P424)	S	S	S	S	N(83)	N(74)	—	—	—	—	—
5L/P75 (P426)	S	S	S	S	N(57)	N(56)	—	—	—	—	—
5L/P76 (P427)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P77 (P428)	0	0	0	0	0	N(56)	—	—	—	—	—
5L/P78 (P429)	0	0	0	S	S	N(73)	—	—	—	—	—
5L/P79 (P432)	0	0	0	0	0	N(61)	—	—	—	—	—
5L/P80 (P433)	0	0	0	0	S*(2395)^{p}	N(46)	—	—	—	—	—
5L/P81	S	S	S	N(64)	N(56)	N(56)	—	—	—	—	—
5L/P82 (P435)	S	S	S	N(47)	N(46)	N(46)	—	—	—	—	—
5L/P83 (P436)	0	0	0	S	N(81)	N(74)	—	—	—	—	—
5L/P84 (P440)	0	0	S	N(46)	N(46)	N(46)	—	—	—	—	—
5L/P85 (P441)	0	0	0	0	0	N(2)	—	—	—	—	—
5L/P86 (P442)	0	0	0	0	S*(2395)^{p}	S*(2395)^{p}	—	—	—	—	—
5L/P87 (P443)	0	0	0	0	0	N(56)	—	—	—	—	—
5L/P88 (P444)	0	0	0	0	0	N(43)	—	—	—	—	—
5L/P89 (P445)	0	0	0	0	0	N(2)	—	—	—	—	—
5L/P90 (P446)	0	0	0	0	0	N(34)	—	—	—	—	—
5L/P91 (P447)	S	S	S	S	S	S	—	—	—	—	—
5L/P92 (P448)	0	0	0	S	N(56)	N(56)	—	—	—	—	—
5L/P93 (P449)	S	S	S	S	S	S	—	—	—	—	—
5L/P94 (P451)	0	0	S	S	N(56)	N(56)	—	—	—	—	—
5L/P95 (P453)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P96 (P454)	0	0	0	0	S*(2600)^{p}	S*(2348)^{p}	—	—	—	—	—
5L/P97 (P455)	0	0	S	N(46)	N(46)	N(46)	—	—	—	—	—
5L/P98 (P456)	0	0	0	0	S*(2600)^{p}	N(46)	—	—	—	—	—
5L/P99 (P457)	0	0	0	0	0	N(52)	—	—	—	—	—
5L/P100 (P460)	0	0	0	S	N(81)	N(74)	—	—	—	—	—
5L/P101 (P462)	S	S	N(47)	N(47)	N(47)	N(47)	—	—	—	—	—
5L/P102 (P463)	0	0	S	N(46)	N(46)	N(46)	—	—	—	—	—
5L/P103 (P464)	0	0	0	0	0	N(47)	—	—	—	—	—
5L/P104 (P465)	0	0	0	0	S*(2600)^{p}	N(46)	—	—	—	—	—
5L/P105 (P466)	0	0	0	0	0	N(54)	—	—	—	—	—
5L/P106 (P470)	S	S	S	S	N(81)	N(73)	—	—	—	—	—

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
5L/P107 (P472)	0	0	0	0	N(46)	N(45)	—	—	—	—	—
5L/P108 (P477)	0	0	0	S	N(56)	N(56)	—	—	—	—	—
5L/P109 (P478)	0	0	0	0	S*(5363)	N(74)	—	—	—	—	—
5L/P110 (P480)	S	S	S	S	N(81)	N(74)	—	—	—	—	—
5L/P111 (P483)	0	0	0	0	0	N(55)	—	—	—	—	—
5L/P112 (P484)	0	0	0	0	0	N(54)	—	—	—	—	—
5L/P113	S	S	S	S	N(55)	N(55)	—	—	—	—	—
5L/P114 (P501)	0	0	0	0	S*(5363)	N(55)	—	—	—	—	—
5L/P115 (P505)	0	0	0	0	0	N(42)	—	—	—	—	—
5L/P116 (P506)	0	0	0	0	S*(5363)	N(56)	—	—	—	—	—
5L/P117 (P509)	0	0	0	0	N(34)	N(34)	—	—	—	—	—
5L/P118 (P511)	0	0	0	0	N(55)	N(56)	—	—	—	—	—
5L/P119 (P519)	0	0	S	S	S*(5527)	S*(5525)	—	—	—	—	—
5L/P120 (P530)	0	0	0	0	S*(5363)	S*(5361)	—	—	—	—	—
5L/P121 (P533)	S	S	S	S	N(81)	N(73)	—	—	—	—	—
5L/P122 (P536)	S	S	S	S	N(80)	N(73)	—	—	—	—	—
5L/P123 (P539)	0	0	0	0	0	N(61)	—	—	—	—	—
5L/P124 (P550)	0	0	0	0	S*(5363)	N(74)	—	—	—	—	—
5L/P125 (P551)	0	0	0	0	0	N(2)	—	—	—	—	—
5L/P126 (P553)	0	0	0	0	0	N(60)	—	—	—	—	—
5L/P127 (P560)	S	S	S	S	N(81)	N(73)	—	—	—	—	—
5L/P128	S	S	S	S	N(55)	N(55)	—	—	—	—	—
5L/P129 (P631)	0	0	0	0	0	N(62)	—	—	—	—	—
5L/P130 (P632)	0	0	0	0	0	N(2)	—	—	—	—	—
5L/P131 (P634)	0	0	0	0	0	N(53)	—	—	—	—	—

[p] Two-profile constants (digits; independent confirmation): vakint::D5bar3 (9924 d; independent implementations agree to 5060 d): 4 orders; vakint::PR0D2Eps1 (2395 d; independent confirmation 48.5 d): 5 orders; vakint::PR0D2Eps2 (2396 d; independent confirmation 51.2 d): 2 orders; vakint::PR12D2Eps2 (2600 d; independent confirmation 48.7 d): 6 orders; vakint::PR12D2Eps3 (2348 d; independent confirmation 421 d): 2 orders; vakint::PR15D2Eps1 (2805 d; independent confirmation 794.27 d): 6 orders; vakint::PR15D2Eps2 (2601 d; independent confirmation 794.42 d): 9 orders; vakint::PR15D2Eps3 (2351 d; independent confirmation 793.39 d): 2 orders; vakint::PR5D2Eps4 (9923 d; independent implementations agree to 5060 d): 4 orders. [n] Numerical mode uses the best payload of a master's coefficient group (same exact basis expression), here a validated numerical value: 4L/P17 ε^0 S*(5363) → 19999 d; 4L/P19 ε^0 S*(5363) → 19999 d; 4L/P52 ε^0 S*(5363) → 19999 d; 4L/P73 ε^0 S*(5363) → 19999 d; 4L/P74 ε^0 S*(5363) → 19999 d.

Precision table 2. The L-loop result — class and digits reachable, worst case over the admitted inputs, per Laurent order of the result, with the spurious-pole orders the row notes state (measured at 1L-4L; at 5L proven on the held integrals and sampled outside them, but not in the top sector: a measured top-sector input is folded in, note k; letters: notes below).

L	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
1L[a]					S	S	S	S	S	S	S
2L[a]				S	S	S	S	S	S	—[b] some inputs: S	—
3L[a]			S	S	S	S	S	—[c] some inputs: S	—[d] some inputs: S	—[e] some inputs: S	—
4L[a]		S	S	S	S[f]	S* 2395 digits[g]	—[h] some inputs: S*(2348)	—[i] some inputs: S	—[j] some inputs: S	—	—
5L[k,l]	S	N 64 digits[m] others: S	N 46 digits[n] others: N(47)	—[o] some inputs: N(46)	—[p] some inputs: N(34)	—[q] some inputs: N(2)	—	—	—	—	—

a 1L-4L: spurious-pole orders p measured on the shipped caches and samples (direct route up to 1L 0, 2L 0, 3L 1, 4L 2), not proven.
b 2L ε^4: inputs reaching 2L/P1 get MissingMasterOrders; others S.
c 3L ε^2: inputs reaching 3L/P1, 3L/P3, 3L/P6 +9 get MissingMasterOrders; others S.
d 3L ε^3: inputs reaching 3L/P1, 3L/P2, 3L/P3 +14 get MissingMasterOrders; others S.
e 3L ε^4: inputs reaching 3L/P1, 3L/P2, 3L/P3 +35 get MissingMasterOrders; only inputs with lower spurious poles, e.g. 3L/P13, 3L/P21, 3L/P28: S.
f 4L ε^{-1} S: the K5Moment(1), K5Moment(3) terms of spurious-pole orders cancel identically (R-operation; exact on all 286527 cache records and 2744 samples).
g 4L ε^0 S*(2395): PR0D2Eps1 (two profiles of one code).
k 5L: p = 0 (pole-free basis BASIS-v2) proven on the 2336 held integrals and sampled outside them. Not so in the top sector: the measured input P105 + dot has p = 1 on 59 5L masters and up to p = 3 on its 4L product factors; other top-sector inputs are not validated.
l 5L results hold products of 1L-4L masters: a factor of L_f loops is needed through ε^{k+5-L_f+p} (p measured on the factor images).
m 5L ε^{-4} N(64): P105 + dot needs 5L/P6 (P60) ε^{-3}; others S.
n 5L ε^{-3} N(46): P105 + dot needs 5L/P50 (P357) ε^{-2}, 5L/P84 (P440) ε^{-2}; others N(47).
o 5L ε^{-2}: P105 + dot gets MissingMasterOrders; others N(46) (5L/P102 (P463) ε^{-2}, 5L/P50 (P357) ε^{-2}.
p 5L ε^{-1}: P105 + dot gets MissingMasterOrders; others N(34) (5L/P117 (P509) ε^{-1}.
q 5L ε^0: P105 + dot gets MissingMasterOrders; others N(2) (5L/P125 (P551) ε^0, 5L/P130 (P632) ε^0.

h 4L ε^1: inputs reaching 4L/P6, 4L/P7, 4L/P9 +18 get MissingMasterOrders; so do 4L/P5, 4L/P8, 4L/P15 +5 combined with 4L/P35, 4L/P51; others S*(2348) (4L/P70 ε^1.
i 4L ε^2: inputs reaching 4L/P1, 4L/P3, 4L/P5 +59 get MissingMasterOrders; others S.
j 4L ε^3: inputs reaching 4L/P1, 4L/P2, 4L/P3 +71 get MissingMasterOrders; only inputs with lower spurious poles, e.g. 4L/P2, 4L/P21, 4L/P37 +1: S.

Method. A coefficient is limited by its fewest-digit finite source (4L constants, numerical master values); the result at ε^k needs every master m through ε^{k+p_m} (p_m: spurious-pole order of its reduction coefficient), on vakint's basis-alias route first, else the direct route; a needed order the data do not hold gives MissingMasterOrders (no value, never a wrong one). RustRed backend only. Full notes, method, finite-digit periods, the numerical identities behind some S and the PSLQ identifications: `crates/vakint/docs/benchmark/data/PRECISION-TABLES.md`.